



Python for Data Science - 2305CS303

Lab - 5

Roll No. : 122

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01) WAP to find sum of all the elements in a List.

```
In [1]: l1=[1,2,3,4,5,6]
        print(sum(l1))
```

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02) WAP to find largest element in a List.

```
In [2]: l1=[1,2,3,4,5,6]
        print(max(l1))
```

6

03) WAP to find the length of a List.

```
In [3]: l1=[1,2,3,4,5,6]
        print(len(l1))
```

6

04) WAP to interchange first and last elements in a list.

```
In [11]: l1=[1,2,3,4,5,6]
         l1[0],l1[len(l1)-1]=l1[len(l1)-1],l1[0]
         print(l1)
```

[6, 2, 3, 4, 5, 1]

05) WAP to split the List into two parts and append the first part to the end.

```
In [13]: l1=[1,2,3,4,5,6]
m=len(l1)//2
l2=l1[:m]
l3=l1[m:]
l3+=l2
l3
```

```
Out[13]: [4, 5, 6, 1, 2, 3]
```

06) WAP to interchange the elements on two positions entered by a user.

```
In [15]: l1=[1,2,3,4,5,6]
n=int(input("enter element index: "))
n1=int(input("enter 2 element index: "))
l1[n],l1[n1]=l1[n1],l1[n]
l1
```

```
Out[15]: [1, 5, 3, 4, 2, 6]
```

07) WAP to reverse the list entered by user.

```
In [39]: l1=[]
n=int(input("enter list length: "))
for i in range(n):
    x=input("enter element: ")
    l1.append(x)
l1.reverse()
l1
```

```
Out[39]: ['8', 'rahul', '5', 'raj', '1']
```

08) WAP to print even numbers in a list.

```
In [19]: l1=[1,2,3,4,5,6]
for i in range(len(l1)):
    if l1[i]%2==0:
        print(l1[i])
```

```
2
4
6
```

09) WAP to count unique items in a list.

```
In [20]: l1=[1,2,3,4,5,6]
s=set(l1)
print(len(s))
```

```
6
```

10) WAP to copy a list.

```
In [26]: from copy import deepcopy
l1=[1,2,3,4,5,6]
l2=l1
l3=deepcopy(l1)
print("l2:",l2)
print("l3: ",l3)
```

```
l2: [1, 2, 3, 4, 5, 6]
l3: [1, 2, 3, 4, 5, 6]
```

11) WAP to print all odd numbers in a given range.

```
In [27]: n=int(input("enter first num: "))
n2=int(input("enter second num: "))
for i in range(n,n2+1):
    if i%2!=0:
        print("odd num is: ",i)
```

```
odd num is:  1
odd num is:  3
odd num is:  5
odd num is:  7
odd num is:  9
```

12) WAP to count occurrences of an element in a list.

```
In [28]: l1=[1,2,3,4,5,6,1]
d={}
for i in l1:
    if i not in d:
        d[i]=1
    else:
        d[i]+=1
print(d)
```

```
{1: 2, 2: 1, 3: 1, 4: 1, 5: 1, 6: 1}
```

13) WAP to find second largest number in a list.

```
In [37]: l1=[1,2,3,4,5,6,1]
l1.sort()
l1[-2]
```

```
Out[37]: 5
```

14) WAP to extract elements with frequency greater than K.

```
In [29]: k=int(input("enter k: "))
l1=[1,2,3,4,5,6,1]
d={}
for i in l1:
    if i not in d:
        d[i]=1
    else:
        d[i]+=1
for j,v in d.items():
```

```
if v==k:  
    print(j)
```

1

15) WAP to create a list of squared numbers from 0 to 9 with and without using List Comprehension.

```
In [31]: s=[i**2 for i in range(0,10)]  
s1=[]  
for i in range(0,10):  
    s1.append(i**2)  
print("s: ",s)  
print("s1: ",s1)
```

```
s: [0, 1, 4, 9, 16, 25, 36, 49, 64, 81]  
s1: [0, 1, 4, 9, 16, 25, 36, 49, 64, 81]
```

16) WAP to create a new list (fruit whose name starts with 'b') from the list of fruits given by user.

```
In [34]: fruit=["apple","banana","grapes","watermelon"]  
fruits=[s for s in fruit if s.lower().startswith('b')]  
fruits
```

```
Out[34]: ['banana']
```

17) WAP to create a list of common elements from given two lists.

```
In [36]: l1=[1,2,3,4,5,6,1]  
l2=[1,2,3]  
s=set(l1) & set(l2)  
list(s)
```

```
Out[36]: [1, 2, 3]
```